

Magic Quadrant for Container Management

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The breadth and depth of container management solutions continue to expand and mature. I&O leaders must navigate a complex marketplace to support agility, modernization and transformation initiatives.

Strategic Planning Assumption

By 2026, the adoption of CSP-native platforms will propel 75% of container instances to be deployed within public cloud environments, up from 50% in 2023.

Market Definition/Description

This is the first version of Magic Quadrant for container management. It replaces the 2022 [Market Guide for Container Management](#).

Gartner defines container management as offerings that enable the deployment and operation of containerized workloads. Delivery methods include cloud, managed service and software.

Container management may be required for containers running on-premises, in the public cloud, at the edge or a combination thereof. Associated technologies include service mesh, orchestration and scheduling, service discovery and registration, image registry, routing and networking, service catalog and management user interface, and API.

Container management is a foundation for cloud-native infrastructure and helps enterprises to automate the provisioning, operation and life cycle management of container images at scale. Centralized governance and security policies manage container instances, clusters and associated resources.

Container management supports the requirements of modern applications, including platform engineering, cloud management and continuous integration/continuous deployment (CI/CD) pipelines. Though challenges remain (e.g., navigating a complex ecosystem and lack of staffing), benefits include improved agility, elasticity and access to innovation. Container management will also assist in support of application modernization where the workload is containerized.

The standard capabilities for container management include:

- **Container infrastructure.** Provides enterprise-grade management of containers across on-premises, cloud and edge locations. Additionally, provides tools for integrated fleet

management, including mind share and configurations for large numbers of container clusters potentially sourced from multiple providers.

- **Managed provisioning and updates.** Automates provisioning and updates of container management functions and infrastructure. This helps enterprises to align their infrastructure with cloud-native concepts and practices.
- **Platform engineering support.** Provides functions needed to deliver a self-service container platform using product management practices.
- **Application platform capabilities.** Provides application development and DevOps functions to accelerate delivery of applications using containers.

Magic Quadrant

Figure 1: Magic Quadrant for Container Management



Vendor Strengths and Cautions

Alibaba Cloud

Alibaba Cloud is a Leader in this Magic Quadrant. Its Alibaba Cloud Container Service for Kubernetes (ACK) container service portfolio is focused on providing a wide range of container deployment options, from single to hybrid and multicloud. Alibaba Cloud's container services are available worldwide, including APAC, Middle East, Western Europe and the United States, but its operations are mostly focused in Asia. Outside of Asia, its clients tend to be organizations running business in China and Chinese multinationals expanding overseas.

Alibaba's Elastic Container Instance (ECI) and Alibaba Cloud Container Service for Kubernetes (ASK) are serverless container services. Alibaba has continually integrated its core public cloud capabilities (e.g., AI) with its container management service.

Strengths

- **Broad Product Portfolio:** Alibaba Cloud has the broadest product portfolio of any non-U.S.-based vendor in this Magic Quadrant. In some use cases, it surpasses U.S.-based leaders. In particular, Alibaba Cloud has advanced products in hybrid cloud and edge solutions through ACK@Edge, Apsara Stack and CloudBox.
- **Alibaba Group's Extensive Use:** Alibaba Cloud's container services are broadly used by Alibaba Group's large e-commerce platform and ecosystem of companies that have demanding requirements, which demonstrates the stability and maturity of Alibaba Cloud's container services.
- **Market Share in China:** Alibaba Cloud has the largest cloud infrastructure as a service (IaaS) market share in Asia, especially in China (where the vast majority of its revenue originates). Alibaba Cloud is growing its presence in the public cloud container service market in China and Southeast Asia by leveraging its large IaaS market share and demonstrating the ability to handle a wide range of user cases.

Cautions

- **Limited Momentum Outside Asia:** Alibaba Cloud has a large IaaS market share in Asia but lacks a presence in other regions. It has yet to become a strong option outside of Asia, except for organizations running business in China and Chinese multinationals expanding overseas.
- **Slowdown in China:** Alibaba Cloud lost 4.5% of its IaaS market share in China in 2022, even though it still holds the top market share in the Chinese market. This slowdown was due to the fines and suspensions from Chinese regulators as well as increased competition.
- **Limited Solution Engineering:** Alibaba Cloud has a rich set of functions, but the solution engineering that brings them together (i.e., concisely with prescriptive clarity) is not as strong as other leaders in this Magic Quadrant.

Amazon Web Services

Amazon Web Services (AWS) is a Leader in this Magic Quadrant. AWS's container service portfolio is focused on supporting a wide range of deployment options, from Kubernetes-based to serverless with a focus on simplicity. Its operations are geographically diversified, and its clients

range from large enterprises to startups. Its customers tend to view AWS as a cloud-native platform, using various AWS services together.

Its primary container services, Amazon Elastic Container Service (Amazon ECS) and Amazon Elastic Kubernetes Service (Amazon EKS), can be used with AWS Fargate (serverless containers). AWS provides a variety of services to complement them, such as container registry, service mesh, observability and security. AWS has been expanding services available outside its public cloud, such as Amazon ECS Anywhere and Amazon EKS Anywhere.

Strengths

- **Integration With AWS and Its Ecosystems:** AWS's container services integrate well with other AWS infrastructure and management services and a wide range of AWS partner offerings on AWS Marketplace. This brings the significant strengths in the core public cloud over to its container management offerings.
- **Customer Base:** AWS's extensive customer list demonstrates that its two main container services, Amazon ECS and Amazon EKS, are highly trusted in production. From startups that value serverless simplicity to complex enterprises that value standards-based governance, a wide variety of customers are running their critical container applications on AWS.
- **Serverless Portfolio:** AWS pioneered serverless container services, releasing AWS Fargate in November 2017. It has continued to expand serverless container options so users, from complex large enterprise organizations to single app development teams, can choose one that suits their requirements.

Cautions

- **Multicloud Support:** AWS does not proactively add features or solutions for managing containers in multicloud environments. Its main multicloud option is to deploy Amazon EKS Distro on other public clouds. This alone is not enough compared to other vendors' multicloud capabilities, and EKS Distro is supported only by AWS partners.
- **Solution Navigation:** AWS has a wide breadth of sometimes overlapping container management offerings. For example, enterprises struggle to differentiate between Amazon EKS and Amazon ECS.
- **Kubernetes Cluster Fleet Management:** AWS publishes best practices for Kubernetes multicluster environments, but lacks native fleet-management tools for managing the life cycle and state of distributed clusters.

Canonical

Canonical is a Niche Player in this Magic Quadrant. Its container management services are focused on providing a wide range of services for diverse compute environments (e.g., enterprise Kubernetes distribution, microdistribution, fleet management tooling and fully managed service). Canonical's operations are geographically diversified, and its clients tend to be early adopters with a focus of modern, agile development.

As a private company, Canonical does not disclose its revenue or customer count. Canonical's most popular offering, Ubuntu Linux, drives much of its product roadmap and engineering effort. Through 2024, Canonical's roadmap focuses on automation, security and maintaining its position in the open-source ecosystem.

Strengths

- **Frictionless Buying:** Pricing is cost-competitive, with all relevant information visible on Canonical's website (including subscriptions, levels of support, training, design/architecture and remote management costs). This provides clarity and simplicity in budgeting and expense management.
- **Edge Computing:** Canonical continues to invest in container management for edge use cases (including MicroK8s, which is a minimal Kubernetes distribution). This makes Canonical suited to test, development and edge use cases.
- **Flexible Deployment:** Canonical's offerings are flexible and particularly suited to multicloud deployment due to Ubuntu's support across many cloud infrastructure providers. Likewise, it is supported on most leading hardware providers and, therefore, equally suited for private and/or hybrid deployment.

Cautions

- **Focus:** Canonical has a wide variety of users within its installed base. However, it has not focused on the needs of traditional enterprises (e.g., requiring a range of infrastructure and application platform services), and its strategy reflects this.
- **Limited Application Mind Share:** Canonical has less traction and distinguishing capabilities with enterprise developers than other vendors in this Magic Quadrant. Because enterprise developers are a primary influencer of container management buying decisions, this puts Canonical at a disadvantage.
- **Business Model:** Public information shows that profitability has been a long-term challenge, even though Canonical's products are in widespread use. Its heavy focus on open-source principles has led to enterprise concerns about vendor viability.

Google

Google is a Leader in this Magic Quadrant. Its Google Cloud Platform is focused on keeping its product portfolio and user experience simple while adding advanced features. Its operations are geographically diversified, and GCP clients range from large enterprises to startups. Google's clients tend to use GCP's container services for cloud-native apps and modern software development.

Google's flagship container management service is Google Kubernetes Engine (GKE). Google Anthos embeds GKE into a broader platform that provisions and manages distributed clusters and services. Anthos can run on-premises, in AWS and Azure as well as in GCP. Google provides serverless containers, such as Google Cloud Run and GKE Autopilot.

Strengths

- **Differentiated Portfolio:** GCP has the highest number of differentiating features of all the vendors' products in this Magic Quadrant. Examples of advanced features include managed service mesh (Anthos Service Mesh), cross-cluster networking capabilities, and tools for managing policy and security across fleets of Kubernetes clusters (Anthos Config Management).
- **Influence Kubernetes Community:** As the creator of Kubernetes and the top vendor by contribution toward Kubernetes, Google has an influential voice in the community and at the forefront in meeting enterprise Kubernetes requirements. Google recently strengthened this position in terms of application platforms by donating Istio and Knative to the Cloud Native Computing Foundation (CNCF).
- **Simplicity:** Google tends to address requirements with fewer services, rather than adding new services, which helps keep its portfolio simple. This principle is also the basis for the simple user experience of Google's serverless container services, such as Google Cloud Run.

Cautions

- **Edge and On-Premises:** GCP has container management products to cover edge and on-premises environments, but it has less adoption and business volume than other leaders in this Magic Quadrant.
- **Traditional Enterprise Systems:** Gartner believes that GCP has a lower presence in traditional enterprise systems use cases such as data center migrations centered on lift and shift or the modernization of legacy applications, compared to cloud-native and modern application development use cases. These traditional systems are where there are many container-based modernization initiatives.
- **Managed Service Provider (MSP) Ecosystem:** GCP lags behind in the MSP ecosystem capable of meeting complex enterprise requirements, when compared to other Leaders in this Magic Quadrant. This includes partners that are driving enterprise application modernization.

Huawei

Huawei is a Challenger in this Magic Quadrant. Its Huawei Cloud container service portfolio is focused on covering broad deployments from on-premises to hybrid cloud and edge environments as well as public cloud. Huawei Cloud's operations are mostly focused on China and emerging countries. Its clients tend to be organizations running business in those countries and Chinese multinationals expanding overseas.

Huawei Cloud's primary Kubernetes managed service is Cloud Container Engine (CCE), which can run on-premises or at the edge, in addition to in the cloud. Huawei Cloud Ubiquitous Cloud Native Service (UCS) provides fleet management that can cover external Kubernetes clusters. Huawei Cloud also has a serverless container service, Cloud Container Instance (CCI).

Strengths

- **Growth in China:** Huawei Cloud is one of the fastest-growing IaaS providers in China. This has culminated in Huawei meeting more complex enterprise requirements. Huawei Cloud leverages its large market share to extend its presence from traditional IaaS-based services into public container services.
- **Contribution to Open Source:** Huawei demonstrates its commitment to open source in this market by contributing to 68 CNCF projects, which is the second highest among the vendors in this Magic Quadrant. Among the open-source projects Huawei donated to CNCF are KubeEdge, Volcano and Karmada.
- **On-Premises and Edge:** Huawei has a mature business for on-premises IT infrastructure and networking offerings that can cover edge use cases. Huawei Cloud leverages these other services to provide container management that can comprehensively cover on-premises, private/hybrid cloud and edge requirements.

Cautions

- **International Sanctions:** International sanctions have a significant impact on Huawei's overall business activities, and organizations may impose strict conditions on sourcing from Huawei Cloud. For this reason, Huawei Cloud does not provide services in some regions, such as the United States.
- **Serverless Options:** Huawei Cloud's serverless container services lack some capabilities offered by other cloud providers in this Magic Quadrant. Its roadmap includes a serverless managed Kubernetes service and a simpler serverless container service for application developers, but they are not yet released.
- **Transparency:** Little information is available related to Huawei Cloud's international offerings. In addition, information about China-based and international services is often mixed and difficult to distinguish from one another.

Microsoft

Microsoft is a Leader in this Magic Quadrant. Its container service portfolio is focused on offering a broad and deep integration with overall services (SaaS, platform as a service [PaaS] and infrastructure as a service [IaaS]), including its application development and DevOps platforms. Its operations are geographically diversified, and its clients range from large enterprises to startups.

Microsoft's primary container service is Azure Kubernetes Service (AKS), which provides various serverless container options, such as AKS virtual node and Azure Container Apps (ACA). Microsoft has been expanding container services that can be used outside of its public cloud, such as AKS on Azure Stack HCI and Azure Arc.

Strengths

- **DevOps Platform:** GitHub and Microsoft Azure DevOps have large mind share and market share as DevOps platforms. Microsoft's container services, which are tightly integrated with those offerings, provide an advantage over other vendors.

- **Hybrid Cloud:** Microsoft offers container service options for hybrid cloud environments. These include AKS, which can be used on Azure Stack HCI, and Azure Arc, which allows users to manage Kubernetes clusters running outside of Azure using Azure Resource Manager.
- **Integration With Microsoft Azure:** Microsoft container management services integrate well with Microsoft Azure and other services, such as Azure Monitor and Azure Security Center. This brings the significant strengths in the core public cloud over to its container management offerings.

Cautions

- **Resiliency:** Gartner clients have reported frustrations with outages of Azure Kubernetes Service (AKS), particularly in association with updates and maintenance events.
- **Heterogeneous Environments Support:** Although Microsoft has an advanced vision of container management in multicloud heterogeneous environments, it still lacks some capabilities, such as full cluster management on other public clouds or VMware vSphere.
- **Certified Container Images:** Microsoft has fewer certified container images for common programming languages and frameworks than other leaders in this Magic Quadrant.

Mirantis

Mirantis is a Niche Player in this Magic Quadrant. Its enterprise container platform includes Mirantis Kubernetes Engine (MKE) as a core element. It also includes Swarm, an early container orchestrator and scheduling offering, which Mirantis acquired from Docker Enterprise in 2019. Its operations are geographically diversified, and many of its clients are in telecommunications and the public sector.

Mirantis continues to invest in developer productivity and reducing operational burden. Its core narrative is associated with ZeroOps, which is focused on automation and streamlined application delivery. Mirantis is a founding member of the OpenStack Foundation, now the OpenInfra Foundation, and remains actively involved. It sponsors and is the primary contributor to the Lens and K0s open-source projects in the CNCF ecosystem and is actively involved in the open-source containerd project with privileged committers to the Moby project.

Strengths

- **Enterprise Track Record:** Mirantis has a track record of supporting enterprise IT infrastructure requirements spanning the last two decades. This provides I&O leaders with support and integration to existing environments (particularly OpenStack).
- **Developer Focus:** Mirantis continues to invest in developer-centric tools (Lens, amaze.io and Shipa). This enables I&O teams to deliver capabilities including an integrated development environment (IDE), DevOps workflow and associated automation.
- **Edge Computing:** Mirantis has developed k0s (a certified Kubernetes distribution deployed as a single binary supporting x86 and ARM processors). This serves the needs of telco and other edge environments.

Cautions

- **Positioning:** Over the last three years, Mirantis has pivoted from its OpenStack heritage to Kubernetes through the Docker Enterprise acquisition. I&O leaders must ensure adequate support, focus and investment for its increasingly hybrid container management needs.
- **Core Functionality:** Technical evaluation of Mirantis' solution shows it to be weak across several Gartner-defined use cases for container management. It lacks differentiators versus others in this evaluation.
- **Customer Traction:** Mirantis has less market traction relative to other vendors in the evaluation. This creates concern about investment relative to its software and cloud-based container management offerings.

Oracle

Oracle is a Niche Player in this Magic Quadrant. Its container service portfolio is focused on security and reliability, which aligns with Oracle Cloud Infrastructure (OCI)'s strengths. Its operations are geographically diversified, and its clients tend to be organizations that use OCI as a strategic IaaS provider.

Oracle's primary container service is Oracle Container Engine for Kubernetes (OKE). It provides a serverless container option on virtual nodes. It enables those services outside its public cloud through Dedicated Region Cloud@Customer and Oracle Private Cloud Appliance (PCA). Oracle also provides Verrazzano, a fleet manager covering on-premises and other public cloud IaaS.

Strengths

- **Private and Sovereign Cloud:** Oracle offers more choice than other cloud providers when it comes to container services on private and sovereign clouds, especially for large enterprise organizations in regulated industries. Examples include OKE on Dedicated Region Cloud@Customer and Kubernetes Engine for Oracle Private Cloud Appliance.
- **Service-Level Agreement (SLA):** Oracle provides an SLA of 99.99% for OKE clusters. This is higher than the typical SLA of 99.95% for most service providers in this Magic Quadrant, making it a good fit for use cases where cluster-level availability is critical.
- **Competitive Price/Performance:** Oracle competes for large opportunities based on price/performance, winning big SaaS vendors and others. OCI's characteristics such as nonblocking networking help achieve better price/performance.

Cautions

- **Developer Experience:** Oracle lacks key features for application platforms and developer experience, such as debugging tools, certified container images and application scaffolding. Additional tools and services need to be integrated to complement these features.
- **Investment Priority for Container Management:** Oracle has been adding container-management features at a good pace, but slower than the Leaders in this Magic Quadrant.

- **Limited, Not Integrated, Fleet Management:** Oracle's fleet manager, Verrazzano, is offered as a stand-alone software product and has its own user interface. This is less convenient and has less functionality than that of other cloud service providers. Its management capabilities for heterogeneous clusters are also limited.

Red Hat

Red Hat is a Leader in this Magic Quadrant. Red Hat OpenShift is broadly focused on container management, with solutions for single and multicloud, on-premises, and edge deployments. Its operations are geographically diversified, and its clients span multiple industries (with strengths in financial services and the public sector).

Red Hat partners with cloud providers to deliver managed Red Hat OpenShift services. These include Red Hat OpenShift Service on AWS (ROSA), Microsoft Azure Red Hat OpenShift (ARO), Red Hat OpenShift on IBM Cloud and Red Hat OpenShift Dedicated on AWS and Google Cloud. It also partners with on-premises infrastructure providers including HPE (GreenLake) and Dell (APEX). Red Hat partners extensively with IBM, its parent company.

Strengths

- **Market Understanding:** Red Hat has developed the product functionality and external partnerships to span the variety of differing container management scenarios. This provides confidence to risk-averse mainstream enterprises looking to adopt container management.
- **Product Capabilities:** Red Hat OpenShift is a strong competitor in the majority of enterprise container management use cases. For I&O leaders seeking a hybrid offering, its pedigree in software-based deployments now extends to distributed cloud and edge computing.
- **Strategic Focus:** Red Hat has a decades-long history in supporting modern enterprise IT infrastructure (with Red Hat OpenStack Platform being an example that Red Hat is starting to incorporate). Its recent acquisitions contribute to the open-source community and also enhance the Red Hat OpenShift platform (e.g., CoreOS, Codenvy, Ansible and StackRox).

Cautions

- **Pricing:** Many Gartner clients have stated that they consider Red Hat OpenShift to be relatively expensive (both container management software subscription and the managed cloud offerings). I&O leaders may struggle to justify the cost for deployments that do not require advanced functionality.
- **Product Dependencies:** Several key features are available only in premium editions but are standard with other vendors (including Red Hat Advanced Cluster Management for Kubernetes). Along with the requirement for a Linux distribution from Red Hat, I&O leaders may be concerned about the degree of lock-in.
- **Competitive Friction:** Red Hat must continue to add application platform functionality to meet customer requirements, which creates tension when it competes with its cloud partners (and

vice versa). I&O leaders will need to decide whether or not to prioritize their strategic cloud vendor's application platform.

SUSE

SUSE is a Challenger in this Magic Quadrant. With its 2020 acquisition of Rancher Labs and history in Linux, SUSE is mainly focused on hybrid and multicloud container management. Its operations are geographically diversified, with a stronghold in Europe. Its container management clients are distributed worldwide.

SUSE has articulated its vision for cloud-native hyperconverged infrastructure, which is known as Harvester. Rancher is also a pioneer in deploying Kubernetes at the edge and in managing heterogeneous Kubernetes distributions.

Strengths

- **Heterogeneity:** Rancher has a history of supporting multiple container management distributions. This allows SUSE customers to avoid lock-in for container workloads and support federated requirements (including autonomous business units).
- **Product Strategy:** The 2020 acquisition of Rancher showcased SUSE's commitment to container management. This provided a foundation for its container-management offering that has expanded over the past few years.
- **Cost-Competitive:** SUSE pricing competes well with the leaders in this evaluation and is often a reason for its selection.

Cautions

- **Stability:** Recent organizational changes follow more than a decade of repeated acquisitions and divestitures of SUSE. This creates concern about longer-term strategy and focus, which is exacerbated by the departure of several of Rancher's most senior executives since late 2021.
- **Differentiation:** SUSE was early to the market. Since then, many other vendors have entered the market with more diverse offerings that better cater to the enterprise application development community.
- **Product Integration:** SUSE is expanding its offering where some of the newer components have not been well-integrated (e.g., NeuVector). This creates concern about fragmentation and a loosely integrated portfolio.

Tencent Cloud

Tencent Cloud is a Challenger in this Magic Quadrant. Its container service portfolio is focused on providing simple, scalable services that empower digital application developers. Tencent Cloud's container services are available worldwide, including Asia, Western Europe, Latin America and North America. Outside of China, its clients tend to be organizations in industries where Tencent Cloud has strengths, such as gaming and consumers.

Tencent Cloud's primary container service is Tencent Kubernetes Engine (TKE) and TKE Serverless. Tencent Kubernetes Engine Distributed Cloud Center (TDCC) provides fleet management that can cover external Kubernetes clusters. TKE has an edge version, TKE Edge, and an open-source version, TKE Stack.

Strengths

- **Gaming and Consumer Industries:** Tencent Cloud has a large market share in the gaming and consumer markets, making it a strong choice for organizations in those industries.
- **Scalability:** Tencent Cloud has IaaS services optimized for high-performance networking and scale-out application architectures. This works as a strength for its container service portfolio and makes Tencent Cloud attractive to customers with high-scale workload requirements.
- **Application Platforms With Digital Ecosystems:** Tencent Cloud's serverless container platform, TCB, is very popular in application development for Tencent Cloud's digital service ecosystems (e.g., games, social networking and digital e-commerce).

Cautions

- **Limited Enterprise Momentum Outside China:** Tencent Cloud lacks a presence for enterprise customers outside China. It has yet to break out of niche markets for gaming, consumer industries, organizations running business in China, and Chinese multinationals expanding overseas.
- **Partner Ecosystems for Enterprises:** Tencent Cloud's partner ecosystems for enterprise businesses outside of China is limited, compared with the Leaders in this Magic Quadrant, from consulting to managed service providers.
- **Smaller Market Share:** Tencent Cloud continues to grow steadily, but still ranks third in terms of market share in China's IaaS market, lagging behind other China-based cloud service providers.

VMware

Note: Broadcom announced its intention to acquire VMware in May 2022. During the evaluation period, VMware met the inclusion criteria for this Critical Capabilities and continued to operate as an independent entity. Gartner will provide additional insight and research as details regarding the acquisition become available.

VMware is a Leader in this Magic Quadrant. Its Tanzu portfolio of container offerings is broadly focused on modern applications. VMware's operations are geographically diversified, and its customers range from midsize to large enterprises. VMware's business model has evolved to embrace container management. It now combines VMware's virtualization products with a focus on modern applications and an increasing contribution to open source.

Broadcom announced its intention to acquire VMware in May 2022. At the publication date, VMware met the inclusion criteria for this Magic Quadrant and continued to operate as an

independent entity. Gartner will provide additional insight and research as details regarding the acquisition become available.

Strengths

- **Customer Base:** VMware has a large installed base and an extensive ecosystem of partners and alliances. It has a successful track record of building on what it already provides, resulting in lesser need to develop new operational skills.
- **Product Strategy:** Ongoing acquisition by VMware demonstrates its commitment to modern applications (e.g., Heptio, Pivotal, Bitnami, Saltstack and Octarine). These acquisitions, combined with ongoing internal R&D investments, highlight how it is adding capabilities to meet the needs of container-based environments.
- **Hybrid Operations:** Leveraging its background in hybrid infrastructure and operational tooling, VMware offers strong capabilities in cluster fleet management across hybrid estates including life cycle management support for public cloud Kubernetes services such as Amazon EKS.

Cautions

- **Competitive Friction:** VMware has built partnerships with leading cloud providers but container management is an area of competitive overlap. To stay broadly competitive in the cloud-native era, VMware must successfully add value that differentiates it from its cloud partner's native container services.
- **Product Integration and Roadmap:** Tanzu comprises container management capabilities originating from multiple internal R&D projects and acquisitions. Gartner clients express confusion at the array of offerings, particularly those delivered as a service, and concern about if/how the different offerings are integrated.
- **Pending Acquisition:** Broadcom's pending acquisition of VMware has been a topic of concern for numerous Gartner clients. Many VMware customers are worried about the potential for price increases, reduced investment in VMware products or impacts to support agreements.

Inclusion and Exclusion Criteria

For Gartner clients, Magic Quadrant and Critical Capabilities research identifies and then analyzes the most relevant providers and products in a market. Gartner uses by default an upper limit of 20 providers to support the identification of the most relevant providers in a market. On some specific occasions, the upper limit may be extended where the intended research value to our clients might otherwise be diminished. The inclusion criteria represent the specific attributes that analysts believe are necessary for inclusion in this research. To qualify for inclusion, providers need to meet the following.

Market Participation Inclusion Criteria

- Provide generally available (GA) capabilities as of 1 April 2023. General availability means the product or service is available on a public-facing price sheet/card for purchase directly by

clients. Providers must be able to furnish the link to a pricing page for their container management services/offerings.

- Sell the solution directly to paying customers without requiring them to engage professional services help. The vendor must provide at least first-line support for these capabilities, including any bundled open-source software. This includes but is not limited to comprehensive product documentation, installation guidance and reference examples.
- Demonstrate an active product roadmap, go-to-market and selling strategy for their solutions.
- Have phone, email and/or web customer support. They must offer contract, console/portal, technical documentation and customer support in English (either as the product's default language or as an optional localization).

Capabilities Inclusion Criteria

The container management offerings must offer native support for the standard capabilities described in Gartner's market definition.

Performance Threshold Achievement

The vendor is required to meet the following financial performance criteria (reported in USD constant currency). The default accounting standard is GAAP.

- The container management offering must have generated at least \$100 million in annual generally accepted accounting principles (GAAP) revenue during the 12 calendar months prior to January 2023. The container management offering must have at least 100 paying production (non-beta-test) customers (excluding sales to managed service providers).

OR

- The container management offering must have generated a minimum of \$30 million in annual revenue and gained 30 net new customers in calendar year 2022 when compared to calendar year 2021.

In addition, the vendor must rank among the top 20 organizations in the market momentum index defined by Gartner for this Magic Quadrant. Data inputs used to calculate container management Magic Quadrant market momentum include a balanced set of measures:

- Gartner customer search, inquiry volume or pricing requests.
- Frequency of mentions as a competitor to other container management Magic Quadrant vendors in reviews for similar use cases on Gartner's Peer Insights forum as of 1 April 2023.
- Scores and frequency of mentions as measured in Gartner Peer Insights.

- Significant innovations in the market as noted by major publications, product enhancements or introductions, or industry awards.
- Other significant developments in corporate posture, such as M&A activity.
- Volume of job listings specifying the container management service/offerings in TalentNeuron’s database or on a range of employment websites in North America, Europe and Asia/Pacific.

Honorable Mentions

IBM: IBM offers a range of container management services (e.g., Red Hat OpenShift on IBM Cloud, IBM Cloud Code Engine, IBM Cloud Satellite, IBM Cloud Kubernetes Service, IBM Cloud Container Registry, and Istio). IBM’s container management strategy is built on Red Hat OpenShift. Red Hat is covered separately and is a Leader in this Magic Quadrant.

Evaluation Criteria

Ability to Execute

Table 1: Ability to Execute Evaluation Criteria

<i>Evaluation Criteria</i> ↓	<i>Weighting</i> ↓
Product or Service	High
Overall Viability	High
Sales Execution/Pricing	Medium
Market Responsiveness/Record	High
Marketing Execution	Medium
Customer Experience	Medium
Operations	Medium

Source: Gartner

Completeness of Vision

Table 2: Completeness of Vision Evaluation Criteria

<i>Evaluation Criteria</i> ↓	<i>Weighting</i> ↓
Market Understanding	High
Marketing Strategy	Medium
Sales Strategy	Medium
Offering (Product) Strategy	High
Business Model	Medium
Vertical/Industry Strategy	Low
Innovation	High
Geographic Strategy	Low

Source: Gartner

Quadrant Descriptions

Leaders

Leaders distinguish themselves by offering a service suitable for strategic adoption and having an ambitious roadmap. They can serve a broad range of use cases, although they do not excel in all areas, may not necessarily be the best providers for a specific need and may not serve some use cases at all. Leaders in this market have appreciable market share and many referenceable customers.

Challengers

Challengers are well-positioned to serve some current market needs. They deliver a good service that is targeted at a particular set of use cases, and they have a track record of successful delivery. However, they are not adapting to market challenges sufficiently quickly or do not have a broad scope of ambition.

Visionaries

Visionaries have an ambitious vision of the future and are making significant investments in the development of unique technologies. Their services are still emerging, and they have many capabilities in development that are not yet generally available. Although they may have many customers, they might not yet serve a broad range of use cases well or may have a limited geographic scope.

Niche Players

The Niche Players in the market for container management may be excellent providers for particular use cases or in regions in which they operate, but they should ultimately be viewed as specialist providers. They often do not serve a broad range of use cases well or have a broadly ambitious roadmap. Some may have solid leadership positions in markets adjacent to this market, but have developed only limited capabilities in this market.

Context

The container management market has seen accelerated growth of 28.6% over the past year with a market value of \$1.6 billion in 2022. The market is forecast to exceed \$3.6 billion in constant currency by 2027, with an 18% CAGR. Among the major container management services are:

- **Container management software.** Solutions that enable the deployment and management of containerized workloads at scale. Some solutions are multifaceted (that is, addressing service mesh, security, observability, etc.); others are primarily Kubernetes distributions. These solutions can be software solutions deployed and managed by enterprise staff or a hired third party (on-premises or within a public cloud environment). This solution can also be offered as a cloud service by a cloud vendor that automates deploying and managing an orchestrator/scheduler (i.e., Kubernetes), but users normally still must manage infrastructure nodes that provide compute resources for containerized workloads.
- **Edge-optimized solutions.** Scaled-down solutions (mostly Kubernetes) that are intended to be deployed at remote locations with minimal service requirements. Increasingly, these solutions combine orchestration and operating system capabilities.
- **Container instance services.** The first generation of serverless container services in which the service automatically provisions container hosts so users do not have to configure or manage such nodes. The underlying container orchestration functionality is not visible to the consumer. Most major cloud service providers offer this type of service.
- **Serverless Kubernetes.** A variant of container instance services that maintains compatibility with Kubernetes applications and APIs. This style automates management of the processing

nodes while allowing users to run Kubernetes pods and use Kubernetes APIs, such as regular Kubernetes services.

- **Cluster management tools.** Solution offerings (often SaaS-based) that enable enterprises to deploy and operate Kubernetes distributions from the same vendor and possibly Kubernetes distributions from other vendors. Their most important features are multicluster life cycle management and “fleet management” of policy, packages and security.

Market Overview

The business and technical benefits associated with container deployments are still mostly centered around enabling enterprise business agility and speed. Recently, more enterprises have commenced container deployments while seeking cost savings and defense against vendor lock-in. Both of these goals are attractive, but difficult to achieve with container deployments. In many cases, container deployments increase costs as enterprises add additional staff and technology assets. And although most solutions are Kubernetes-based, vendors add other components that make the entire technology stack proprietary, which often prevents portability, thus enforcing lock-in.

Many long-standing challenges continue today (for example, operational complexity, security concerns and tooling immaturity). Many enterprises are still early in this journey.

Container-related expertise is still limited, but improving. Many enterprises are aggregating support within a platform engineering/operations group that sits between the application development groups and organizations providing infrastructure (that is, I&O and/or cloud providers). Such groups prove to be efficient versus having support distributed among multiple application development groups.

Two new challenges are enterprises commencing container deployments without an ability to adopt DevOps principles and practices and enterprises taking a “build it and they will come” approach. Both approaches often lead to disappointment with container deployments. The combination of DevOps and container technology can be a powerful enabler for application development agility and speed, making DevOps skills the critical factor to deployment success. And I&O should never deploy container management without deep knowledge of developer requirements.

The most impactful deployment approach is to leverage containers to build net new applications. This allows development efforts to be unencumbered by legacy systems. Many enterprises have targeted the refactoring of existing applications. Examination of application portfolios is also increasing attention on the volume of applications that can be optimized or refactored.

Interest in hybrid and multicloud use cases is growing. The broader public cloud focus has evolved into a hybrid (in some cases, distributed) and/or multicloud discussion in which most enterprises are open to using multiple cloud environments. Many enterprises believe that containers and/or Kubernetes enable multicloud. However, although most vendors offer Kubernetes services, each tends to have proprietary features that make them “sticky,” inhibiting easy portability. Management solutions offer the ability to operationalize management across

these different cloud environments through a multienvironment control panel. Additionally, public cloud vendors are offering container management software that can be deployed in a variety of environments. These offerings initially targeted more of a hybrid than multicloud model (that is, extending their technology from the cloud to on-premises).

Gartner sees some bare-metal deployments, but the rate of growth is slow. This approach is often seen as a way to replace hypervisors. A major drawback continues to be a lack of operational tooling to support bare-metal deployments.

Finally, vendors are continually enhancing their marketplaces to enable easy access to third-party container tools and applications that are optimized for containers and Kubernetes, sometimes packaged as operators that help to manage their life cycle. Closely monitoring a vendor's activity in this domain is a good way to gauge the health of its ecosystem.

Evidence

This research is based on more than 2,000 customer interactions over the past 12 months. In addition, as part of our analysis, we have collected information from Gartner Peer Insights, client inquiries and publicly available sources to supplement the information provided by participating vendors.

Notes

For Alibaba Cloud, Huawei Cloud, and Tencent Cloud: This Magic Quadrant evaluation is focused on their international businesses, and our technical assessment was performed using their international services.

Evaluation Criteria Definitions

Ability to Execute

Product/Service: Core goods and services offered by the vendor for the defined market. This includes current product/service capabilities, quality, feature sets, skills and so on, whether offered natively or through OEM agreements/partnerships as defined in the market definition and detailed in the subcriteria.

Overall Viability: Viability includes an assessment of the overall organization's financial health, the financial and practical success of the business unit, and the likelihood that the individual business unit will continue investing in the product, will continue offering the product and will advance the state of the art within the organization's portfolio of products.

Sales Execution/Pricing: The vendor's capabilities in all presales activities and the structure that supports them. This includes deal management, pricing and negotiation, presales support, and the overall effectiveness of the sales channel.

Market Responsiveness/Record: Ability to respond, change direction, be flexible and achieve competitive success as opportunities develop, competitors act, customer needs evolve and market dynamics change. This criterion also considers the vendor's history of responsiveness.

Marketing Execution: The clarity, quality, creativity and efficacy of programs designed to deliver the organization's message to influence the market, promote the brand and business, increase awareness of the products, and establish a positive identification with the product/brand and organization in the minds of buyers. This "mind share" can be driven by a combination of publicity, promotional initiatives, thought leadership, word of mouth and sales activities.

Customer Experience: Relationships, products and services/programs that enable clients to be successful with the products evaluated. Specifically, this includes the ways customers receive technical support or account support. This can also include ancillary tools, customer support programs (and the quality thereof), availability of user groups, service-level agreements and so on.

Operations: The ability of the organization to meet its goals and commitments. Factors include the quality of the organizational structure, including skills, experiences, programs, systems and other vehicles that enable the organization to operate effectively and efficiently on an ongoing basis.

Completeness of Vision

Market Understanding: Ability of the vendor to understand buyers' wants and needs and to translate those into products and services. Vendors that show the highest degree of vision listen to and understand buyers' wants and needs, and can shape or enhance those with their added vision.

Marketing Strategy: A clear, differentiated set of messages consistently communicated throughout the organization and externalized through the website, advertising, customer programs and positioning statements.

Sales Strategy: The strategy for selling products that uses the appropriate network of direct and indirect sales, marketing, service, and communication affiliates that extend the scope and depth of market reach, skills, expertise, technologies, services and the customer base.

Offering (Product) Strategy: The vendor's approach to product development and delivery that emphasizes differentiation, functionality, methodology and feature sets as they map to current and future requirements.

Business Model: The soundness and logic of the vendor's underlying business proposition.

Vertical/Industry Strategy: The vendor's strategy to direct resources, skills and offerings to meet the specific needs of individual market segments, including vertical markets.

Innovation: Direct, related, complementary and synergistic layouts of resources, expertise or capital for investment, consolidation, defensive or pre-emptive purposes.

Geographic Strategy: The vendor's strategy to direct resources, skills and offerings to meet the specific needs of geographies outside the "home" or native geography, either directly or through partners, channels and subsidiaries as appropriate for that geography and market.

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